

POLITECNICO
MILANO 1863

Requirement Analysis and Specification Document, Luca Suzani, Alberto Spezzapria, Alessio Pesavento

Requirement Analysis and Specification Document

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Authors:	Luca Suzani, Alberto Spezzapria, Alessio Pesavento
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Chapter 1

Introduction

1.1 Purpose

The purpose of the *Best Bike Paths (BBP)* system is to promote cycling as a sustainable, safe, and conscious means of transportation by providing cyclists with a digital platform for collecting, accessing, and sharing information about cycling routes and activities.
The system addresses the need for reliable and up-to-date data on bike paths, allowing users to track their performance while contributing to the improvement of shared information.
Through active community participation, BBP aims to identify and maintain the safest and most efficient routes, fostering a more informed and widespread cycling mobility.
Ultimately, BBP seeks to enhance the overall cycling experience by integrating multiple functionalities and creating a collaborative, continuously updated information network driven by user contributions.

1.1.1 Goals

G1. **Support cyclists in tracking their activities:** Enable users to record their cycling routes and access detailed statistics such as total distance, average speed, and other performance indicators, enriched, when available, by meteorological data from external services (weather conditions, temperature and wind speed).

G2. **Allow cyclists to contribute information about bike paths:** Provide two modes for entering information related to bike paths:

- **Manual mode:** the user manually inputs the path data and its condition.
- **Automatic mode:** the system automatically collects data via GPS, accelerometer, and gyroscope during the cycling activity, requiring user confirmation before publication.

G3. **Enable the publication of recorded path information:** Allow users to make their recorded path information public, contributing to a shared and valuable database for the entire cycling community.

G4. **Enable route search and visualization:** Allow users to search for and display on a map one or more cycling routes connecting a chosen origin and destination, ranking them based on overall condition and effectiveness in reaching the goal.

G5. **Maintain up-to-date and reliable path information:** Ensure continuous updating of path data through the aggregation and analysis of user feedback, taking into account data freshness, consistency among reports, and user confirmations.

1.2 Scope

The main goal of the *Best Bike Paths (BBP)* system is to provide cyclists with a mobile application that allows them to record their activities and share information about bike routes, while also fostering the creation of an active and collaborative community. The application enables registered users to save their rides, view detailed statistics such as total distance covered or average speed, and contribute to updating the path inventory by reporting the condition of routes or the presence of potential obstacles. The information entered by users can be made public and is then aggregated by the system, which combines data from multiple sources to offer a comprehensive and up-to-date representation of the paths' conditions. Unregistered users can still take advantage of the consultation features by searching for routes between two points and viewing the available options on a map, ranked according to a score that considers both the quality and effectiveness of the routes. The system also interacts with an external meteorological service, which provides weather data to enrich the information related to recorded activities.

1.2.1 World Phenomena

1.2.2 Shared Phenomena

1.3 Definitions, Acronyms, and Abbreviations

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

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Chapter 2

Overall Description

Here you can see how to include an image in your document.

Here is the command to refer to another element (section, figure, table, ...) in the document: *As discussed in Section 2 and as shown in Figure 2.1, ...* Here is how to introduce a bibliographic citation [**?**]. Bibliographic references should be included in a `.bib` file.

Table generation is a bit complicated in Latex. You will soon become proficient, but to start you can rely on tools or external services. See for instance this https://www.tablesgenerator.com.

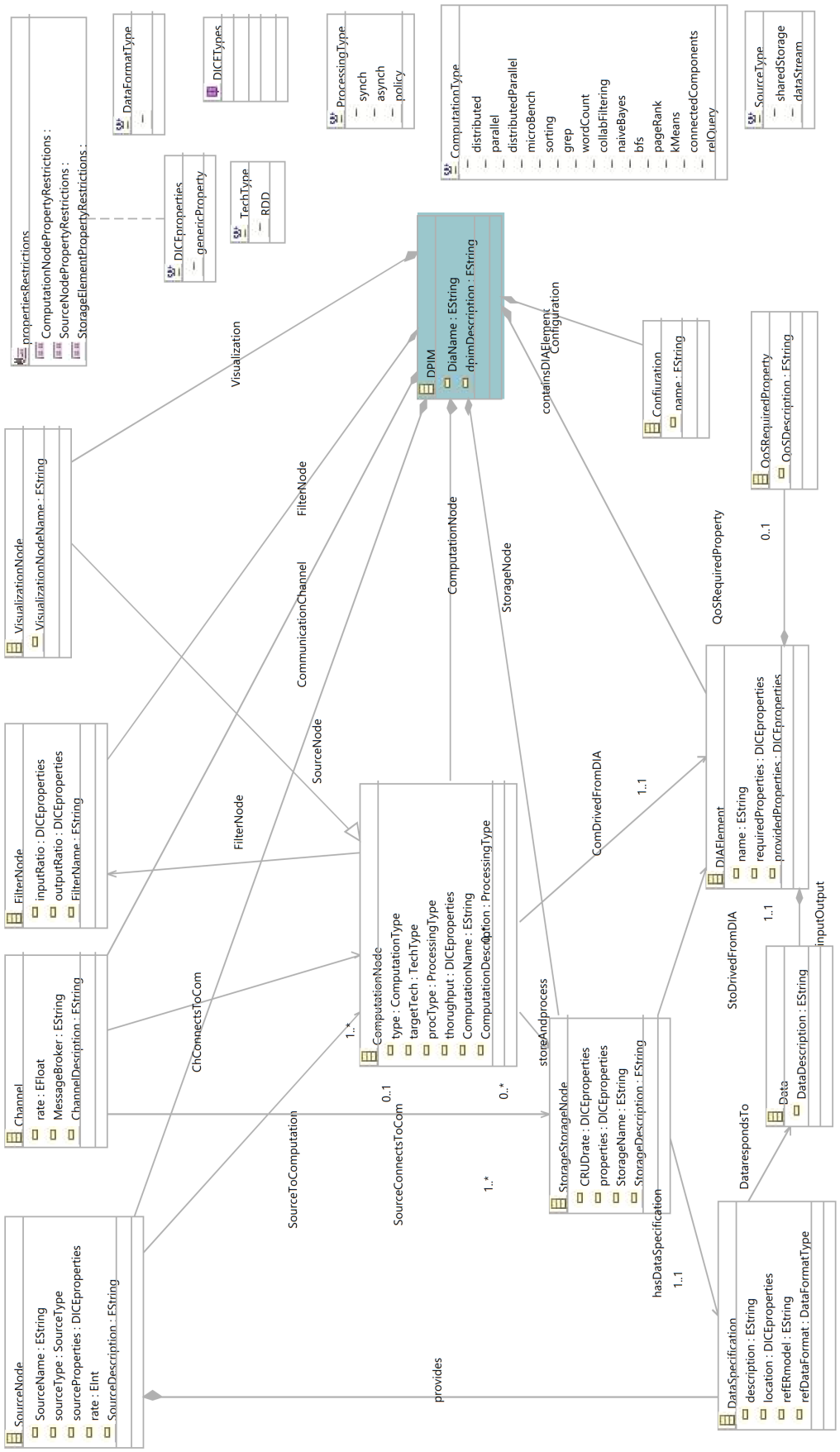


Figure 2.1: DICE DPIM metamodel.

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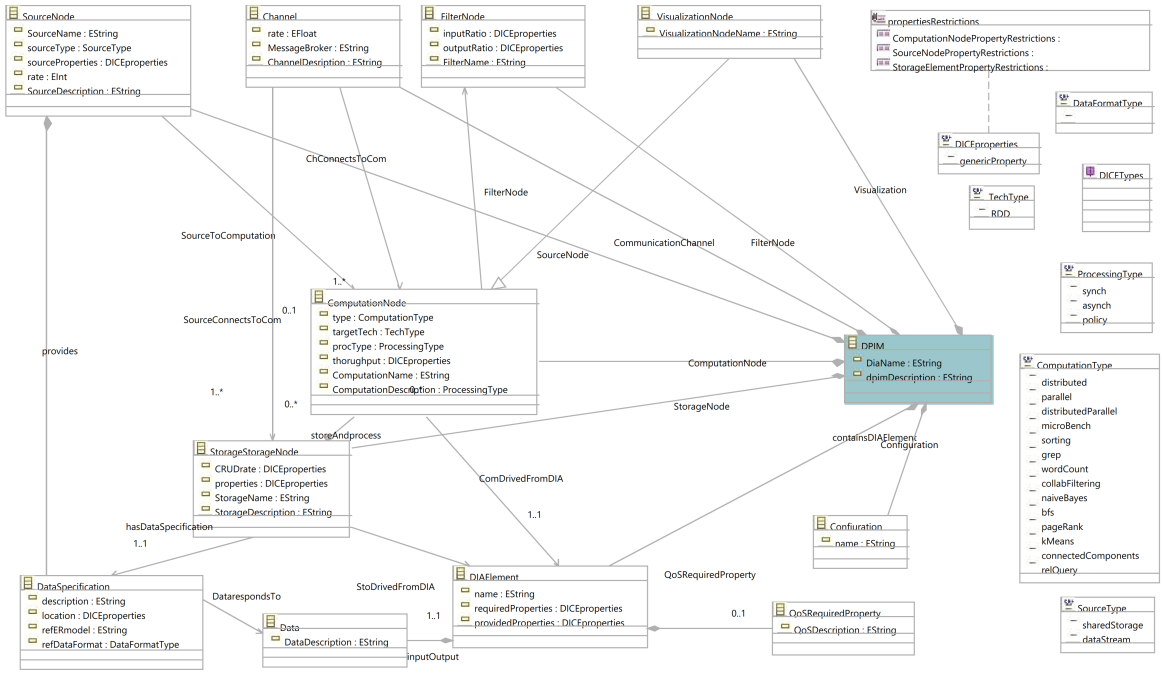


Figure 2.2: DICE DPIM metamodel in portrait form.

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Chapter 3

Specific Requirements

Organize this section according to the rules defined in the project description.

Chapter 4

Formal Analysis Using Alloy

Organize this section according to the rules defined in the project description.

Chapter 5

Effort Spent

Data	Alessio Pesavento	Alberto Spezzapria	Luca Suzani
27/10/2025	1h	1h	1h
07/11/2025	2h	2h	2h
08/11/2025	-	1h	-
09/11/2025	1h	-	-
10/11/2025	-	-	1h
11/11/2025	-	-	-

Table 5.1: Hours of work spent by each author on each day on the RASD document